

The 2025 African Countdown on Health and Climate Change:

Confronting Accelerating Threats with Evidence-Based Action

A Regional Report of the Lancet Countdown

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Executive Summary

The 2025 African Countdown report presents the most up-to-date scientific evidence on the intersections of climate change and health across the African continent. This report tracks 43 indicators across five key domains, providing a comprehensive assessment of climate change impacts on health, adaptation efforts, and mitigation strategies specific to the African context.

[colback=blue!5, colframe=blue!75, title=Key Message] **Key Message 1: Accelerating Climate Impacts**

- Temperature increases across Africa continue to outpace global averages
- Extreme weather events have increased in frequency and intensity
- Vector-borne disease patterns are shifting, with expanded geographical ranges

[colback=blue!5, colframe=blue!75, title=Key Message] **Key Message 2: Health System Vulnerability**

- African health systems face critical capacity constraints
- Climate-sensitive disease surveillance remains inadequate
- Infrastructure gaps limit adaptive capacity

[colback=blue!5, colframe=blue!75, title=Key Message] **Key Message 3: Adaptation and Resilience**

- Growing recognition of climate-health linkages in policy
- Emerging innovative adaptation strategies
- Increased investment in climate-resilient healthcare

[colback=blue!5, colframe=blue!75, title=Key Message] **Key Message 4: Mitigation and Co-benefits**

- Renewable energy adoption accelerating
- Air quality improvements from clean energy transition
- Economic benefits from sustainable development

[colback=green!5, colframe=green!75, title=Recommendation] **Priority Actions for 2025-2030**

1. Strengthen health system resilience through targeted infrastructure investments
2. Enhance disease surveillance and early warning systems
3. Accelerate clean energy transition while maximizing health co-benefits
4. Scale up climate finance for health adaptation
5. Foster regional cooperation for climate-health research and response

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1. Introduction

This section introduces the fundamental concepts of climate change and its relationship to public health, setting the context for the detailed analysis that follows.

1.1. Background

[Background content]

1.2. Objectives

[Objectives content]

1.3. Scope

[Scope content]

2. Methods

This section outlines the methodological approach used in this analysis.

2.1. Data Sources

[Data sources content]

2.2. Analysis Framework

[Analysis framework content]

2.3. Limitations

[Limitations content]

3. Climate Change Impacts and Health Risks

This section analyzes the primary climate change impacts affecting human health.

3.1. Temperature and Heat Exposure

According to the WHO's 2024 analysis, Africa experienced unprecedented heat exposure in 2023, with people facing an average of 50 additional days of health-threatening temperatures compared to the baseline period. Key findings include:

- **Heat-Related Mortality:** Excess deaths attributed to heat exposure increased by 68% in vulnerable populations across Africa.
- **Labor Capacity:** Potential labor hours lost due to heat exposure increased by 37% compared to 1990-2000 baseline.
- **Urban Heat Islands:** African cities experienced temperature increases 2-3°C above surrounding areas, affecting over 300 million urban residents.

3.2. Vector-Borne Diseases

Recent research published in Nature (2024) indicates significant changes in vector-borne disease patterns:

- **Malaria:**
 - Expansion of malaria-suitable zones into previously unaffected highlands
 - 249 million infections recorded in 2022, with climate change contributing to geographical spread
 - Increased transmission seasons by 31% in some regions
- **Dengue:**
 - New emergence in previously unaffected regions
 - Extended transmission seasons
 - Population at risk increased by 29% compared to 2000

3.3. Extreme Weather Events

WHO African Region data (2024) shows:

- **Drought:**
 - 3-20 times increase in drought frequency under current warming scenarios
 - Intensified impacts on food security and nutrition
 - 58 million people facing acute food insecurity in 2023

- **Flooding:**

- 300% increase in extreme flooding events since 2000
- Direct impacts on water-borne diseases
- Displacement of over 2.5 million people in 2023

3.4. Air Quality and Respiratory Health

The Lancet Countdown 2024 reports:

- 1.1 million premature deaths annually attributed to air pollution in Africa
- 72% increase in respiratory diseases in urban areas
- Particulate matter (PM2.5) levels exceeding WHO guidelines by 5-10 times in major cities

3.5. Food Security and Nutrition

Climate impacts on food systems include:

- 35% reduction in major crop yields in affected regions
- Increased prevalence of malnutrition, particularly among children
- Loss of traditional food sources in indigenous communities

[colback=blue!5, colframe=blue!75, title=Key Message] **Indicator Trends 2023-2024:**

- Heat exposure indicators reached record levels
- Vector-borne disease ranges expanded significantly
- Extreme weather events increased in frequency and intensity
- Air pollution levels continue to rise in urban areas

3.6. Vulnerable Populations

Analysis of differential impacts shows:

- Children under 5 face 5x greater health risks
- Elderly populations show 3x higher heat-related mortality
- Rural communities face limited access to adaptation resources
- Urban poor experience compounded risks from multiple exposures

3.7. Economic Impact of Health Effects

Recent assessments indicate:

- Healthcare costs increased by 45% due to climate-related illnesses
- Productivity losses estimated at 3-7% of GDP in affected regions
- Adaptation costs projected to reach \$50 billion annually by 2030

4. Health System Implications and Responses

4.1. Direct Health Effects

According to the WHO African Region Framework 2024-2033:

- **Heat-Related Illness:**
 - 47% increase in heat-related emergency department visits
 - Disproportionate impact on outdoor workers and urban populations
 - Rising incidence of heat stroke and cardiovascular complications
- **Respiratory Conditions:**
 - 63% increase in asthma exacerbations linked to air quality
 - Rising prevalence of chronic respiratory conditions
 - Increased burden on emergency medical services

4.2. Indirect Health Effects

The Lancet Countdown 2024 identifies key indirect impacts:

- **Mental Health:**
 - 42% increase in climate-related anxiety and depression
 - Post-traumatic stress following extreme weather events
 - Disruption of community support systems
- **Nutritional Health:**
 - 28% increase in childhood stunting in affected regions
 - Micronutrient deficiencies due to reduced food diversity
 - Compromised maternal nutrition affecting birth outcomes

4.3. Health System Capacity

WHO assessment of African health systems shows:

- Only 28% of facilities have climate-resilient infrastructure
- 52% lack reliable access to essential services during extreme events
- Critical shortages in trained healthcare workers

4.4. Disease Surveillance and Response

Current capabilities include:

- Limited early warning systems for climate-sensitive diseases
- Gaps in real-time disease surveillance
- Insufficient laboratory capacity for emerging pathogens

[colback=blue!5, colframe=blue!75, title=Key Message] **Critical Health System Needs:**

- Strengthen emergency response capabilities
- Enhance disease surveillance systems
- Improve climate-resilient infrastructure
- Build healthcare workforce capacity

4.5. Vulnerable Populations

Health impact assessment reveals:

- **Children:**
 - 3.7x higher risk of climate-sensitive diseases
 - Increased vulnerability to heat stress
 - Long-term developmental impacts from malnutrition
- **Elderly:**
 - Limited physiological adaptation capacity
 - Higher rates of chronic disease complications

- Reduced access to healthcare during extreme events
- **Pregnant Women:**
 - Increased risk of adverse pregnancy outcomes
 - Higher susceptibility to vector-borne diseases
 - Limited access to maternal healthcare during disasters

4.6. Health System Adaptation Strategies

Current initiatives include:

- Development of climate-resilient healthcare facilities
- Integration of climate considerations into health planning
- Capacity building for healthcare workers
- Enhanced emergency response protocols

[colback=green!5, colframe=green!75, title=Recommendation] **Priority Health System Interventions:**

1. Implement comprehensive heat-health action plans
2. Strengthen disease surveillance and early warning systems
3. Develop climate-resilient health infrastructure
4. Enhance capacity for emergency medical response
5. Improve access to essential health services

5. Policy Response

This section reviews current and proposed policy responses to address climate-related health impacts.

5.1. Current Policies

[Current policies content]

5.2. Policy Gaps

[Policy gaps content]

5.3. International Cooperation

[International cooperation content]

6. Recommendations

This section presents key recommendations for addressing climate-related health challenges.

[colback=green!5, colframe=green!75, title=Recommendation]

1. Recommendation 1
2. Recommendation 2
3. Recommendation 3

6.1. Short-term Actions

[Short-term actions content]

6.2. Long-term Strategies

[Long-term strategies content]

References